

Overview in General

This file contains a general overview of the data in the graph including node labels and relationships types.

References

- [jqassistant](#)
- [Neo4j Python Driver](#)

Node Labels

Table 1a - Highest node count by label combination

Lists the 30 label combinations with the highest number of nodes. The labels with the lowest node count are listed in table 1b. The total list would sum up to the total number of labels (100%).

The whole table can be found in the CSV report `Node_label_combination_count`.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Git, Update, Change]	138637	46.084372
1	[Git, Change, Create]	44924	14.933202
2	[Git, Change, Delete]	22780	7.572308
3	[Java, ByteCode, Method, Member]	13482	4.481556
4	[Java, ByteCode, Parameter]	13293	4.418731
5	[Git, Commit]	11339	3.769201
6	[File, Git]	8285	2.754020
7	[Java, ByteCode, Bound, ParameterizedType]	7279	2.419615
8	[Java, ByteCode, Bound]	7240	2.406651
9	[Git, Change, Rename]	7018	2.332856
10	[Java, ByteCode, Member, Field]	3605	1.198339
11	[Java, ByteCode, Bound, WildcardType]	2947	0.979613
12	[Java, Value, ByteCode, Annotation]	2931	0.974295
13	[Xml, Element]	2162	0.718671
14	[Java, ByteCode, Constructor, Method, Member]	2113	0.702383
15	[Xml, Text]	1450	0.481995
16	[Java, ByteCode, TypeVariable, Bound]	1111	0.369308
17	[Java, ByteCode, Method, Member, Lambda]	968	0.321773
18	[Type, File, Java, ByteCode, ResolvedDuplicate...	888	0.295180
19	[Type, File, Java, ByteCode, Class]	846	0.281219
20	[Json, Key]	694	0.230693
21	[Value, Json, Scalar]	678	0.225374
22	[Java, Value, ByteCode, Primitive]	672	0.223380
23	[Type, File, Java, ByteCode, JavaType]	649	0.215734
24	[Java, ByteCode, Method, Member, GenericDeclar...	578	0.192133
25	[Type, File, Java, ByteCode, ExternalType]	411	0.136621
26	[Git, Change, Copy]	298	0.099058
27	[Author, Git, Person]	295	0.098061
28	[Value, Array]	287	0.095402
29	[Committer, Git, Person]	242	0.080443

Chart 1a - Highest node count by label combination

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in Chart 1b.

<Figure size 640x480 with 0 Axes>

Nodes per label combination (more than 0.5% overall)

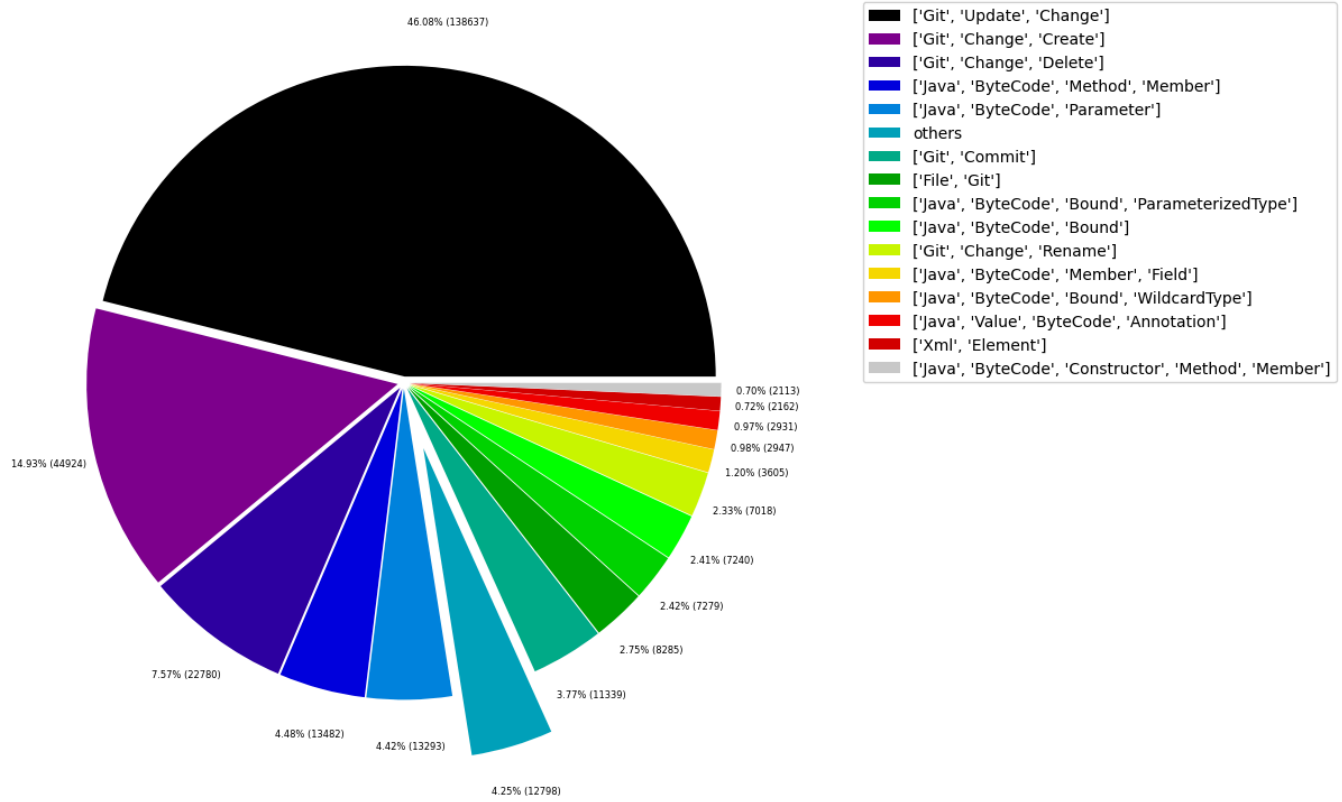


Table 1b - Lowest node count by label combination

Lists the 30 label combinations with the lowest number of nodes until they reach 0.5% of the total node count, which are shown above.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Analyze, Task, jQAssistant]	1	0.000332
1	[Git, Branch]	1	0.000332
2	[Repository, File, Git]	1	0.000332
3	[File, Json]	2	0.000665
4	[File]	3	0.000997
5	[Java, ByteCode, Constructor, Method, Member, ...]	4	0.001330
6	[Maven, Exclusion]	5	0.001662
7	[Value, Array, Json]	6	0.001994
8	[Type, File, Java, ByteCode, Void]	9	0.002992
9	[File, Maven, Xml, Pom, Document]	9	0.002992
10	[Java, ManifestSection]	9	0.002992
11	[File, Artifact, Jar, Archive, Zip, Java]	9	0.002992
12	[File, Java, Manifest]	9	0.002992
13	[File, Java, ServiceLoader]	10	0.003324
14	[File, Java, Properties]	12	0.003989
15	[Maven, ExecutionGoal]	16	0.005319
16	[Maven, PluginExecution]	16	0.005319
17	[Xml, Attribute]	18	0.005983
18	[jQAssistant, Rule, Concept]	19	0.006316
19	[Maven, Plugin]	21	0.006981
20	[Maven, Configuration]	21	0.006981
21	[Type, File, Java, ByteCode, Throwable, Extern...	21	0.006981
22	[Type, File, Java, ByteCode, Throwable, Resolv...	22	0.007313
23	[Type, File, Java, ByteCode, Enum]	28	0.009307
24	[Type, File, Java, ByteCode, PrimitiveType]	30	0.009972
25	[Xml, Namespace]	36	0.011967
26	[Type, File, Java, ByteCode, Annotation]	44	0.014626
27	[File, Directory]	44	0.014626
28	[Type, File, Java, ByteCode, Class, Throwable]	55	0.018283
29	[Type, File, Java, ByteCode, GenericDeclaratio...	85	0.028255

Chart 1b - Lowest node count by label combination

Shows the lowest (less than 0.5% overall) node count label combinations. Therefore, this plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

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Nodes per label combination (less than 0.5% overall)

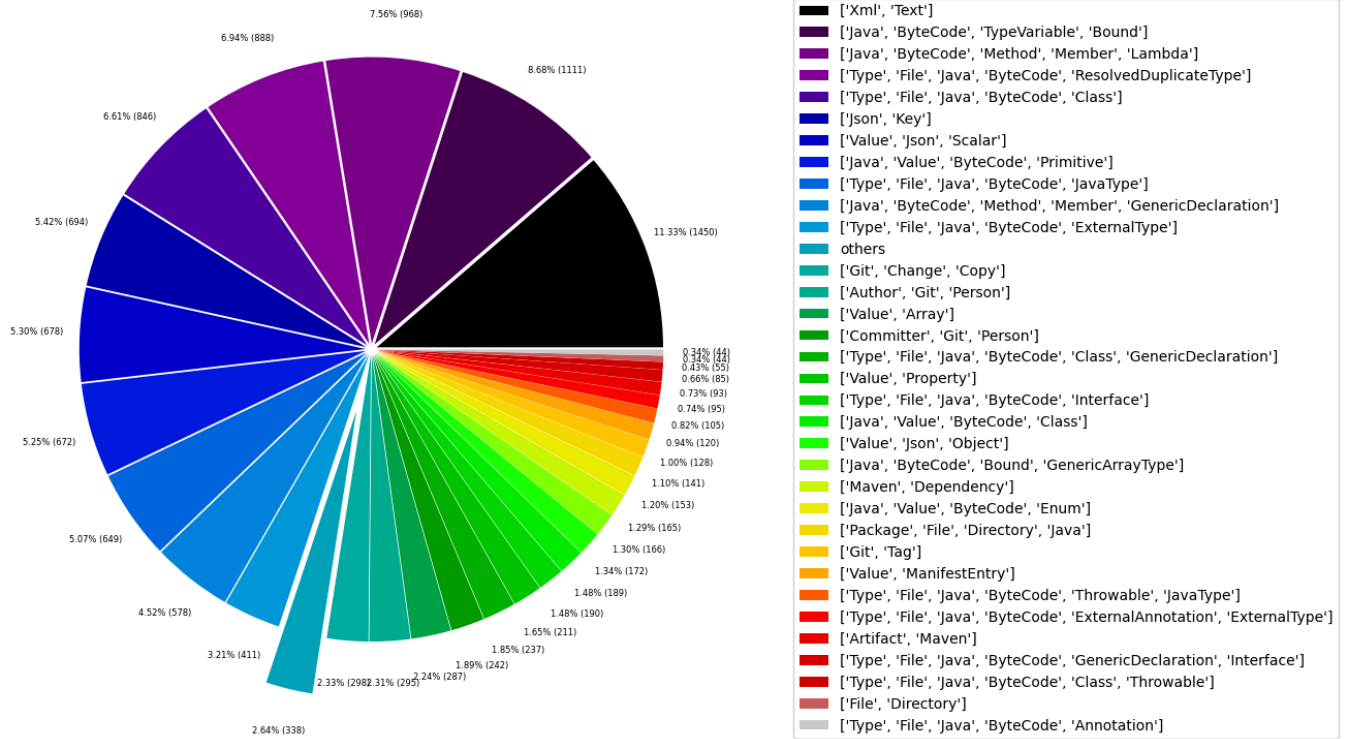


Table 1c - Highest node count by single label

Lists the 40 labels with the highest number of nodes. Doesn't sum up to the total number of nodes or 100% because one node can have multiple labels. Helps to identify commonly used labels.

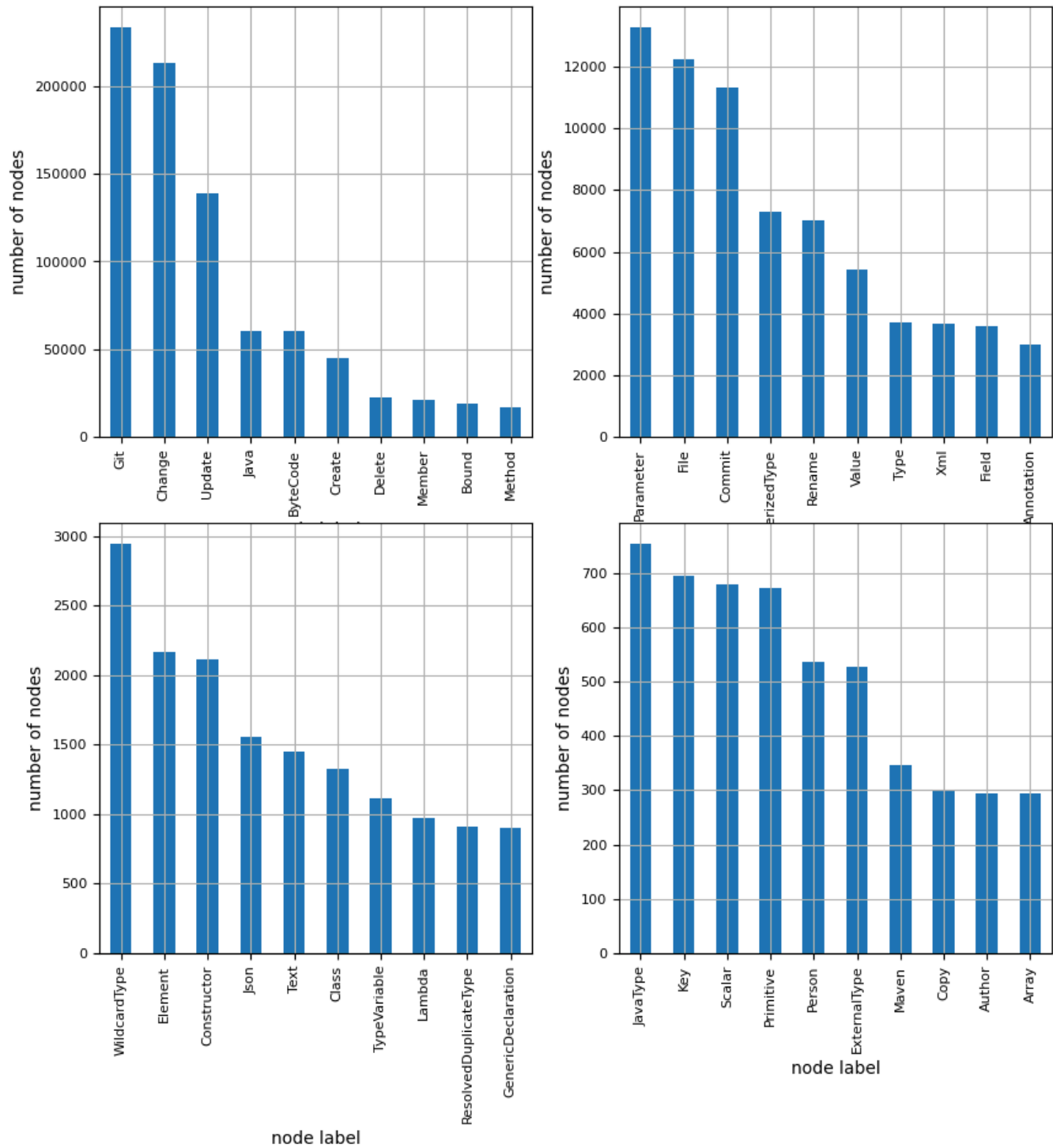
	nodeLabel	nodesWithThatLabel	nodesWithThatLabelPercent
0	Git	233948	77.766734
1	Change	213657	71.021796
2	Update	138637	46.084372
3	Java	60636	20.156033
4	ByteCode	60446	20.092875
5	Create	44924	14.933202
6	Delete	22780	7.572308
7	Member	20750	6.897515
8	Bound	18743	6.230367
9	Method	17145	5.699175
10	Parameter	13293	4.418731
11	File	12240	4.068703
12	Commit	11339	3.769201
13	ParameterizedType	7279	2.419615
14	Rename	7018	2.332856
15	Value	5419	1.801332
16	Type	3715	1.234904
17	Xml	3675	1.221608
18	Field	3605	1.198339
19	Annotation	2975	0.988921
20	WildcardType	2947	0.979613
21	Element	2162	0.718671
22	Constructor	2117	0.703713
23	Json	1552	0.515901
24	Text	1450	0.481995
25	Class	1327	0.441109
26	TypeVariable	1111	0.369308
27	Lambda	968	0.321773
28	ResolvedDuplicateType	910	0.302493
29	GenericDeclaration	904	0.300499
30	JavaType	754	0.250637
31	Key	694	0.230693
32	Scalar	678	0.225374
33	Primitive	672	0.223380
34	Person	537	0.178504
35	ExternalType	527	0.175180
36	Maven	346	0.115014
37	Copy	298	0.099058
38	Author	295	0.098061
39	Array	293	0.097396

Chart 1c - Highest node count by label

Shows the 40 labels with the highest number of nodes.

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Node count by label



Relationship Types

Table 2a - Highest relationship count by type

Lists the 30 relationship types with the highest number of occurrences. The whole table can be found in the CSV report `Relationship_type_count`.

Total number of relationships: 935189

	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	CONTAINS_CHANGE	213657	22.846398
1	MODIFIES	213657	22.846398
2	UPDATES	138637	14.824490
3	CREATES	52240	5.586037
4	INVOKES	36786	3.933536
5	DELETES	29798	3.186308
6	COMMITTED	22678	2.424964
7	DEPENDS_ON	22492	2.405075
8	OF_TYPE	21920	2.343911
9	DECLARES	21211	2.268098
10	OF_RAW_TYPE	17353	1.855561
11	HAS	14429	1.542897
12	HAS_PARENT	13858	1.481839
13	RETURNS	12844	1.373412
14	HAS_COMMIT	11339	1.212482
15	READS	9379	1.002899
16	HAS_ACTUAL_TYPE_ARGUMENT	8407	0.898963
17	HAS_FILE	8285	0.885917
18	RENAMES	7018	0.750437
19	OF_GENERIC_TYPE	5996	0.641154
20	RESOLVES_TO	4234	0.452743
21	SIMILAR	4078	0.436062
22	WRITES	3944	0.421733
23	CONTAINS	3926	0.419808
24	HAS_NEW_NAME	3662	0.391579
25	RETURNS_GENERIC	3593	0.384200
26	CHANGED_TOGETHER_WITH	3027	0.323678
27	ANNOTATED_BY	2919	0.312129
28	REQUIRES	2230	0.238454
29	HAS_FIRST_CHILD	2162	0.231183

Chart 2a - Highest relationship count by type

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in the second chart.

<Figure size 640x480 with 0 Axes>

Relationship types (more than 0.5% overall)

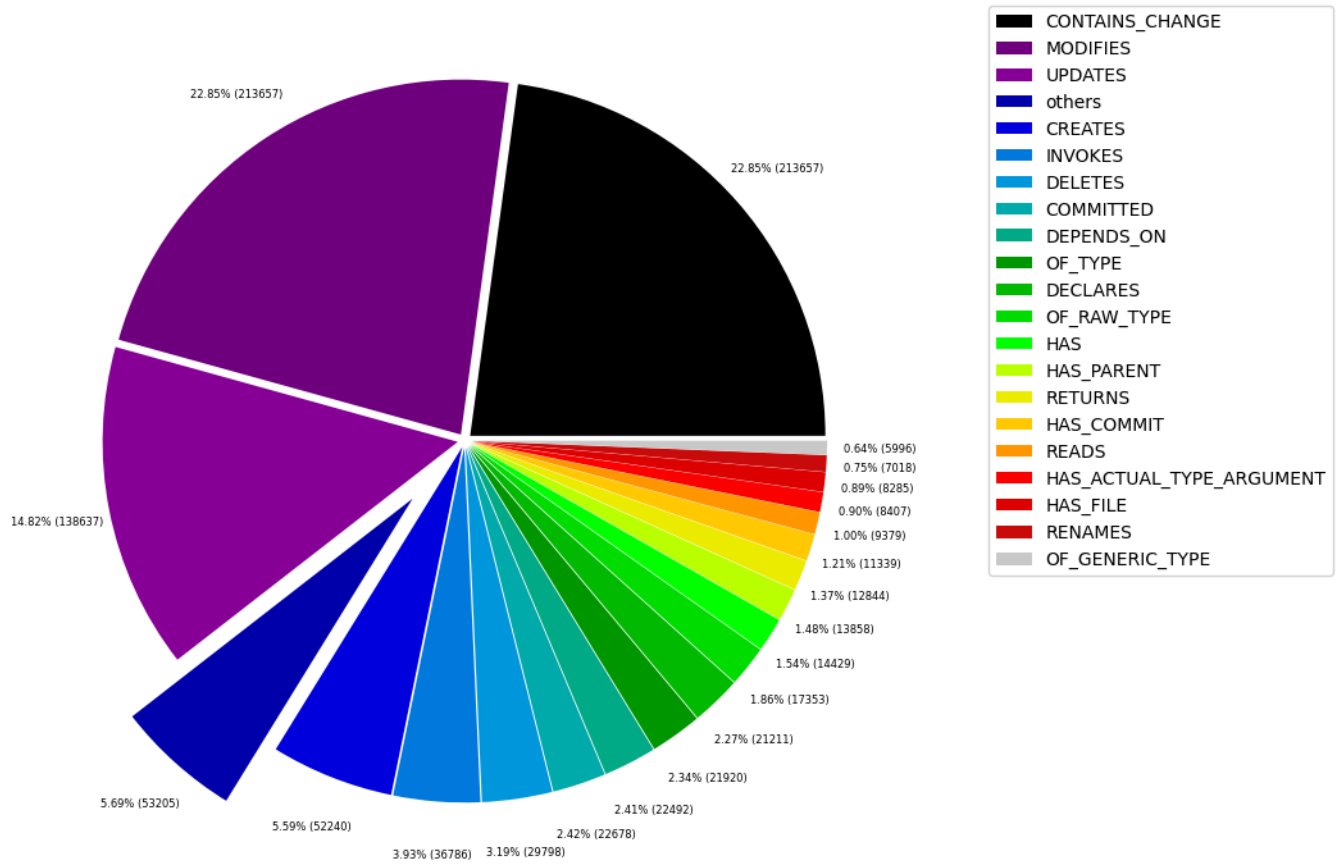


Table 2b - Lowest relationship count by type

Lists the 30 relationships type with the lowest number of occurrences up to 0.5% of the total node count. This is essentially breaking down the "others" slice from the chart above.

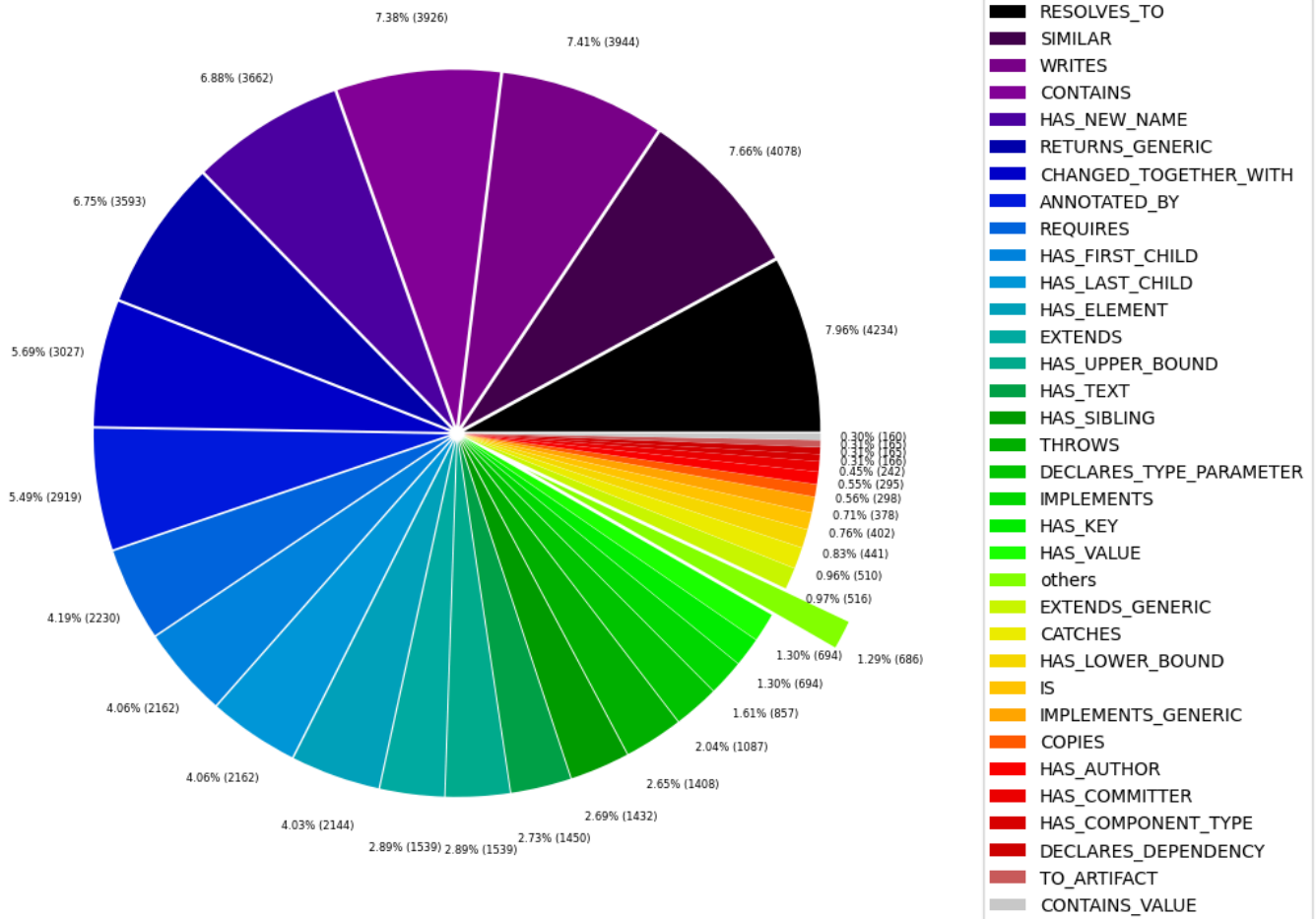
	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	HAS_PROPERTY	1	0.000107
1	HAS_BRANCH	1	0.000107
2	HAS_HEAD	2	0.000214
3	THROWS_GENERIC	5	0.000535
4	EXCLUDES	5	0.000535
5	DESCRIBES	9	0.000962
6	HAS_ROOT_ELEMENT	11	0.001176
7	HAS_GOAL	16	0.001711
8	HAS_EXECUTION	16	0.001711
9	OF_NAMESPACE	18	0.001925
10	HAS_ATTRIBUTE	18	0.001925
11	INCLUDES_CONCEPT	19	0.002032
12	USES_PLUGIN	21	0.002246
13	IS_ARTIFACT	21	0.002246
14	HAS_CONFIGURATION	21	0.002246
15	REQUIRES_TYPE_PARAMETER	24	0.002566
16	REQUIRES_CONCEPT	28	0.002994
17	DECLARES_NAMESPACE	36	0.003849
18	HAS_DEFAULT	39	0.004170
19	COPY_OF	119	0.012725
20	HAS_TAG	128	0.013687
21	ON_COMMIT	128	0.013687
22	CONTAINS_VALUE	160	0.017109
23	TO_ARTIFACT	165	0.017643
24	DECLARES_DEPENDENCY	165	0.017643
25	HAS_COMPONENT_TYPE	166	0.017750
26	HAS_COMMITTER	242	0.025877
27	HAS_AUTHOR	295	0.031544
28	COPIES	298	0.031865
29	IMPLEMENTS_GENERIC	378	0.040420

Chart 2b - Lowest relationship count by type

Shows the lowest (less than 0.5% overall) relationship types. This plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

<Figure size 640x480 with 0 Axes>

Relationship types (less than 0.5% overall)



Node labels with their relationships

Table 3a - Highest relationship count by node labels and relationship type

Lists the 30 node labels and their relationship types with the highest number of occurrences.

	sourceLabels	relationType	targetLabels	numberOfRelationships	numberOfNodesWithSameLabelsAsSource	numbe
0	[Git, Update, Change]	UPDATES	[File, Git]	138637	138637	
1	[Git, Update, Change]	MODIFIES	[File, Git]	138637	138637	
2	[Git, Commit]	CONTAINS_CHANGE	[Git, Update, Change]	138637	11339	
3	[Git, Change, Create]	CREATES	[File, Git]	44924	44924	
4	[Git, Change, Create]	MODIFIES	[File, Git]	44924	44924	
5	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Create]	44924	11339	
6	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Delete]	22780	11339	
7	[Git, Change, Delete]	MODIFIES	[File, Git]	22780	22780	
8	[Git, Change, Delete]	DELETES	[File, Git]	22780	22780	
9	[Java, ByteCode, Method, Member]	INVOKES	[Java, ByteCode, Method, Member]	22484	13382	
10	[Git, Commit]	HAS_PARENT	[Git, Commit]	13849	11339	
11	[Repository, File, Git]	HAS_COMMIT	[Git, Commit]	11339	1	
12	[Author, Git, Person]	COMMITTED	[Git, Commit]	11339	295	
13	[Committer, Git, Person]	COMMITTED	[Git, Commit]	11339	242	
14	[Java, ByteCode, Method, Member]	HAS	[Java, ByteCode, Parameter]	8506	13382	
15	[Java, ByteCode, Method, Member]	READS	[Java, ByteCode, Member, Field]	8444	13382	
16	[Repository, File, Git]	HAS_FILE	[File, Git]	8285	1	
17	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Rename]	7018	11339	
18	[Git, Change, Rename]	CREATES	[File, Git]	7018	7018	
19	[Git, Change, Rename]	MODIFIES	[File, Git]	7018	7018	
20	[Git, Change, Rename]	DELETES	[File, Git]	7018	7018	
21	[Git, Change, Rename]	RENAMES	[File, Git]	7018	7018	
22	[Java, ByteCode, Parameter]	OF_TYPE	[Type, File, Java, ByteCode, JavaType]	6156	13293	
23	[File, Git]	HAS_NEW_NAME	[File, Git]	3662	8285	
24	[Java, ByteCode, Bound]	OF_RAW_TYPE	[Type, File, Java, ByteCode, JavaType]	3465	7240	
25	[Java, ByteCode, Bound, ParameterizedType]	OF_RAW_TYPE	[Type, File, Java, ByteCode, JavaType]	3110	7279	
26	[Java, ByteCode, Bound, ParameterizedType]	HAS_ACTUAL_TYPE_ARGUMENT	[Java, ByteCode, Bound, WildcardType]	2947	7279	
27	[Java, ByteCode, Parameter]	OF_GENERIC_TYPE	[Java, ByteCode, Bound, ParameterizedType]	2695	13293	
28	[Java, ByteCode, Constructor, Method, Member]	WRITES	[Java, ByteCode, Member, Field]	2519	2113	
29	[Java, ByteCode, Bound, ParameterizedType]	HAS_ACTUAL_TYPE_ARGUMENT	[Java, ByteCode, TypeVariable, Bound]	2474	7279	

Graph Density

```
total_number_of_nodes (vertices): 300833
total_number_of_relationships (edges): 935189
-> total directed graph density: 1.0333558066791313e-05
-> total directed graph density in percent: 0.0010333558066791314
```